MINISTERUL EDUCAȚIEI
UNIVERSITATEA TEHNICĂ
DIN CLUJ-NAPOCA

PROIECT

FUNDAȚII

Proiectarea fundațiilor izolate pentru stâlpii unei structuri în cadre

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BORDEROU

A. Parte scrisă

- Foaie de prezentare
- Tema proiectului
- Memoriu tehnic
- Note de calcul

B. Parte desenată

- Plan fundații: scara 1:50
- Detalii caracteristice: scara 1:20 – plan și secțiune S1,S2,S3

MEMORIU TEHNIC

I. INTRODUCERE

- Proiectul urmărește dimensionarea și verificarea fundațiilor izolate pentru stâlpii unei structuri în cadre. Studiul este realizat conform normativelor în vigoare, respectând prevederile SR EN 1997-1 și principiile de calcul geotehnic și structural.

II. OBIECTIVELE PROIECTULUI ȘI VERIFICĂRI NECESARE

- Dimensionarea fundațiilor pentru stâlpi de colț, marginali și centrali (S1, S2, S3);
- Stabilitatea generală și locală a fundațiilor la SLU și SLS conform SR EN 1997-1.
- Calculul tasărilor totale și diferențiale pentru a respecta limitele admise.
- Verificarea presiunii admisibile a terenului de fundare.
- Alegerea unei soluții optime din punct de vedere tehnic și economic.

III. DESCRIEREA GENERALĂ A STRUCTURII ȘI AMPLASAMENTULUI

- Structura este compusă din cadre cu stâlpi din beton armat, având o secțiune dreptunghiulară de 400x500 mm. Pereții de închidere sunt din zidărie BCA de 30 cm grosime, iar grinzile sunt de beton armat.

TEREN DE FUNDARE:

- **Strat superior** (0-0,50 m): Umplutură, $\gamma = 17,2$ kN/m³.
- **Strat intermediar** (0,50-3,50 m): Argilă prafoasă, consistență plastic consistentă, $\gamma = 18,7$ kN/m³, $w = 22\%$, $c = 29$ kPa.
- **Strat inferior** (3,50-10,00 m): Nisip argilos, $\gamma = 19,3$ kN/m³, $\varphi = 31°$.

COTA TERENULUI SISTEMATIZAT (C.T.S.): -0,70 m.

TIPURI DE FUNDAȚII PROIECTATE

- **S1 (stâlp de colț):** Fundație rigidă izolată (FIR).
- **S2 (stâlp marginal):** Fundație rigidă izolată (FIR).
- **S3 (stâlp central):** Fundație elastică izolată (FIE).

MATERIALE UTILIZATE: Beton: C16/20, Oțel S500

CONCLUZII

- Soluțiile de fundații propuse asigură cerințele de rezistență, stabilitate și utilizare impuse de normativ.
- Se recomandă respectarea detaliilor constructive specificate în planurile de execuție și realizarea unor verificări suplimentare la fața locului înaintea execuției lucrărilor.

TEMA 3
FUNDAȚII IZOLATE

DATE PROIECT

$n := 4.5$
$a := 15\ cm$
$H_z := 3\ m$
$C.T.S := -(0.55\ m + a) = -0.7\ m$

Stratificația terenului

Umplutură

$h_1 := 0.5\ m$

$\gamma_{k1} := 17.2\ \frac{kN}{m^3}$

Argilă prăfoasă cafenie, plastic consistentă

$h_2 := 3\ m$

$\gamma_{k2} := 18.7\ \frac{kN}{m^3}$

$w_2 := 22\%$

$I_{p2} := 22\%$

$I_{c2} := 0.70$

$\phi'_{k2} := 15\ deg$

$e_2 := 0.63$

$E_2 := 28.000\ kPa$

$c'_{k2} := 29.45\ kPa$

Nisip argilos galben cafeniu

$h_3 := 6.5\ m$

$\gamma_{k3} := 19.3\ \frac{kN}{m^3}$

$w := 18\%$

$I_{p3} := 11\%$

$\phi'_{k3} := 31\ deg$

$e_3 := 0.6$

$E_3 := 42.000\ kPa$

$c'_{k3} := 9.25\ kPa$

Încărcări de calcul pentru fiecare stâlp

- încărcarea distribuită permanentă: $P_k := 16.4\ kPa$

- încărcarea distribuită variabilă: $Q_k := 7.4\ kPa$

NOTE DE CALCUL

$\gamma_G := 1.35$ $\quad P_d := P_k \cdot \gamma_G = 22.14\ kPa$ $\quad H := 3\ m$ $\quad l_{st} := 0.5\ m$

$\gamma_Q := 1.35$ $\quad Q_d := Q_k \cdot \gamma_Q = 9.99\ kPa$ $\quad b_z := 0.3\ m$ $\quad b_{st} := 0.4\ m$

PREDIMENSIONARE STÂLP S1 (stâlp de colț) + S2 (stâlp marginal)

Eforturi secționale

$A_{S2} := 0.5\ m \cdot 0.4\ m = 0.2\ m^2$

$N_{S2} := A_{S2} \cdot \left(P_d + Q_d\right) = 6.426\ kN$

$M_{x2} := 0.2\ m \cdot N_{S2} = 1.285\ kN \cdot m$

$M_{y2} := 0.1\ m \cdot N_{S2} = 0.643\ kN \cdot m$

$T_{x2} := 0.1 \cdot N_{S2} = 0.643\ kN$

$T_{y2} := 0.05 \cdot N_{S2} = 0.321\ kN$

$H_i := 0.90\ m$ $\quad D_{f2} := H_i + 20\ cm = 1.1\ m$

Dimensiunea tălpii fundației: L și B

$P_{eff} \le P_{adm}$

$P_{conv} := 297.25\ kPa$ $\quad P_{adm} := 1.2\ P_{conv} = 356.7\ kPa$

$V_{d2} := 1.2\ N_{S2} = 7.711\ kN$

$r_2 := \dfrac{l_{st}}{b_{st}} = 1.25$

$\dfrac{1.2 \cdot N}{B_2 \cdot L_2} \le P_{adm}$ $\quad \dfrac{1.2\ N_{S2}}{{B_2}^2 \cdot r_2} \le P_{adm}$ $\quad B_{2min} := \sqrt{\left(\dfrac{V_{d2}}{r_2 \cdot P_{adm}}\right)} = 0.132\ m$

$B_2 := 2\ m$

$L_2 := r_2 \cdot B_2 = 2.5\ m$

Dimensiunea tălpii cuzinetului: lc și bc

$l_{c2min} := 0.5 \cdot L_2 = 1.25\ m$ alegem $l_{c2} := 1.25\ m$

$b_{c2min} := 0.5 \cdot B_2 = 1\ m$ alegem $b_{c2} := 1.00\ m$

Înălțimea blocului de fundare și a cuzinetului: H și hc

$tan\alpha \geq tan\alpha_{adm}$ $tan\alpha_{adm} := 1.3$

$H_2 := tan\alpha_{adm} \cdot \frac{(L_2 - l_{c2})}{2} = 0.813\ m$ $tan\alpha := \frac{H_2}{\frac{L_2 - l_{c2}}{2}} = 1.3$

$H_2 := tan\alpha_{adm} \cdot \frac{(B_2 - b_{c2})}{2} = 0.65\ m$ $tan\alpha := \frac{H_2}{\frac{B_2 - b_{c2}}{2}} = 1.3$

alegem $H_2 := 0.85\ m$

$tan\beta \geq tan\beta_{adm}$

$tan\beta_{adm} := 1$ $tan\beta := tan\beta_{adm}$

$h_{c2} := tan\beta \cdot \frac{(l_{c2} - l_{st})}{2} = 0.375\ m$ $tan\beta_2 := \frac{l_{c2}}{\frac{(l_{c2} - l_{st})}{2}} = 3.333$

$h_{c2} := tan\beta \cdot \frac{(b_{c2} - b_{st})}{2} = 0.3\ m$ $tan\beta_2 := \frac{b_{c2}}{\frac{(b_{c2} - b_{st})}{2}} = 3.333$

alegem $h_{c2} := 0.40\ m$

$\frac{l_{c2}}{h_{c2}} \geq 2.5$ **VERIFICĂ**

VERIFICAREA LA CAPACITATE PORTANTĂ A TĂLPII FUNDAȚIEI

$P_{eff} \leq P_{adm}$

$\frac{V_d}{A'} \leq \frac{R_d}{A'}$

Efortul vertical de calcul la talpa fundației

$L_{af2} := 6\ m$

$\gamma_{BCA} := 16\ \frac{kN}{m^3}$ $\gamma_{b.a} := 24\ \frac{kN}{m^3}$ $\gamma_{med} := 22\ \frac{kN}{m^3}$

$b_{gr} := b_z = 0.3\ m$

$1.5 \leq \frac{h_{gr}}{b_{gr}} \leq 3$ $h_{gr} := 0.85\ m$ $\frac{h_{gr}}{b_{gr}} = 2.833$ **VERIFICĂ**

$H_{tot} := H_2 + h_{c2} + h_{gr} = 2.1\ m$

$N_{cd2} := N_{S2} = 6.426\ kN$

$G_z := b_z \cdot H \cdot L_{af2} \cdot \gamma_{BCA} \cdot \gamma_G = 116.64\ kN$

$G_{f2} := B_2 \cdot L_2 \cdot H_{tot} \cdot \gamma_{med} \cdot \gamma_G = 311.85\ kN$

$G_{gr} := b_{gr} \cdot h_{gr} \cdot L_{af2} \cdot \gamma_{b.a} \cdot \gamma_G = 49.572\ kN$

$V_{d2} := N_{cd2} + G_{f2} + G_{gr} + G_z = 484.488\ kN$

Momentele încovoietoare la talpa fundației

$M_{xf2} := M_{x2} + T_{y2} \cdot (h_{c2} + H_2) = 1.687\ kN \cdot m$

$M_{yf2} := M_{y2} + T_{x2} \cdot (h_{c2} + H_2) + (G_z + G_{gr}) \cdot \left(\frac{b_{st}}{2} - \frac{b_z}{2}\right) = 9.756\ kN \cdot m$

Forțele orizontale de la talpa fundației

$T_{xf2} := T_{x2} = 0.643\ kN$

$T_{yf2} := T_{y2} = 0.321\ kN$

$H_{d2} := \sqrt{T_{xf2}{}^2 + T_{yf2}{}^2} = 0.718\ kN$

Calculul presiunii efective la talpa fundației

$e_B := \frac{M_{yf2}}{V_{d2}} = 0.02\ m$

$e_L := \frac{M_{xf2}}{V_{d2}} = 0.003\ m$

VERIFICAREA EXCENTRICITĂȚII

$\left(\frac{e_B}{B_2}\right)^2 + \left(\frac{e_L}{L_2}\right) \le \frac{1}{9} = 1$ **VERIFICĂ**

$B'_2 := B_2 - 2 \cdot e_B = 1.96\ m$

$L'_2 := L_2 - 2\ e_L = 2.493\ m$

$A'_2 := B'_2 \cdot L'_2 = 4.886\ m^2$

$\gamma_{c'} := 1.25 \qquad \gamma_\gamma := 1$

$c'_{d2} := \frac{c'_{k2}}{\gamma_{c'}} = 23.56\ \frac{kN}{m^2}$

$\gamma' := \frac{\gamma_{k2}}{\gamma_\gamma} = 18.7\ \frac{kN}{m^3}$

$q' := D_{f2} \cdot \gamma' = 20.57\ \frac{kN}{m^2}$

$\tan\left(\phi'_{k2}\right) = 0.268$

$N_q := e^{\pi \cdot \tan\left(\phi'_{k2}\right)} \cdot \left(\tan\left(45\ ° + \frac{\phi'_{k2}}{2}\right)\right)^2 = 3.941 \qquad N_q := 3.941$

$$N_c := (N_q - 1) \cdot \frac{1}{\tan(\phi'_{k2})} = 10.976 \qquad m_B := \frac{2 + \frac{B'_2}{L'_2}}{1 + \frac{B'_2}{L'_2}} = 1.56$$

$$N_\gamma := 2\ (N_q - 1) \cdot \tan(\phi'_{k2}) = 1.576 \qquad m_L := \frac{2 + \frac{L'_2}{B'_2}}{1 + \frac{L'_2}{B'_2}} = 1.44$$

$$b_q := 1 \qquad b_\gamma := 1$$

$$b_c := b_q - \frac{(1 - b_q)}{N_c \cdot \tan(\phi'_{k2})} = 1 \qquad \theta := \operatorname{atan}\left(\frac{T_{y2}}{T_{x2}}\right) = 26.565\ ^\circ$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_2}{L'_2}\right) = 0.764$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_2}{L'_2}\right) = 0.764$$

$$s_q := 1 + \frac{B'_2}{L'_2} \cdot \sin(\phi'_{k2}) = 1.203 \qquad m_\theta := m_L \cdot \left(\cos(\theta)^2 + m_B \cdot \sin(\theta)^2\right) = 1.601$$

$$s_c := \frac{s_q \cdot N_q - 1}{N_q - 1} = 1.273$$

$$i_q := \left(1 - \left(\frac{0.5}{550 + 1.338 \cdot 23.52 \cdot \cos(\phi'_{k2})}\right)\right)^{m_\theta} = 0.999$$

$$i_\gamma := \left(1 - \left(\frac{0.5}{550 + 1.338 \cdot 29.4 \cdot \cos(\phi'_{k2})}\right)\right)^{m_\theta} = 0.999$$

$$i_c := i_q - \frac{(1 - i_q)}{N_c \cdot \tan(\phi'_{k2})} = 0.998 \qquad P_{adm2} := \frac{R_d}{A'_2}$$

$$P_{adm2} := c'_{d2} \cdot N_c \cdot b_c \cdot s_c \cdot i_c + q' \cdot N_q \cdot b_q \cdot s_q \cdot i_q + 0.5 \cdot \gamma' \cdot B'_2 \cdot N_\gamma \cdot b_\gamma \cdot s_\gamma \cdot i_\gamma = 447.95\ \frac{kN}{m^2}$$

$$\frac{V_{d2}}{A'_2} = 99.165\ \frac{kN}{m^2} \qquad P_{adm2} > \frac{V_{d2}}{A'_2}$$

VERIFICĂ

ARMAREA CUZINETULUI

$b_{c2} := 1.00\ m$ $\quad L_{af2} = 6\ m$ $\quad \gamma_{bca} := 16\ \frac{kN}{m^3}$

$l_{c2} := 1.25\ m$ $\quad h_{gr} = 0.85\ m$ $\quad \gamma b.a. := 24\ \frac{kN}{m^3}$

$h_{c2} := 0.4\ m$ $\quad b_{gr} = 0.3\ m$ $\quad \gamma_{med} := 22\ \frac{kN}{m^3}$

$H_{tot.c2} := h_{gr} + h_{c2} = 1.25\ m$

$M_{xc2} := M_{x2} + T_{y2} \cdot h_{c2} = 1.414\ kN \cdot m$

$M_{yc2} := M_{y2} + T_{x2} \cdot h_{c2} = 0.9\ kN \cdot m$

$G_{gr} := b_{gr} \cdot h_{gr} \cdot L_{af2} \cdot \gamma_{bca} \cdot \gamma_G = 33.048\ kN$

$G_z := b_z \cdot H \cdot L_{af2} \cdot \gamma_{bca} \cdot \gamma_G = 116.64\ kN$

$G_{c2} := H_{tot.c2} \cdot b_{c2} \cdot l_{c2} \cdot \gamma_{med} \cdot \gamma_G = 46.406\ kN$

$N_{c2} := N_{S2} + G_{gr} + G_z + G_{c2} = 202.52\ kN$

$A_{c2} := b_{c2} \cdot l_{c2} = 1.25\ m^2$

$$P_{1c} := \frac{N_{c2}}{b_{c2} \cdot l_{c2}} + 6 \cdot \frac{M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 167.445\ \frac{kN}{m^2}$$

$$P_{2c} := \frac{N_{c2}}{l_{c2} \cdot b_{c2}} - \frac{6 \cdot M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 156.588\ \frac{kN}{m^2}$$

$$P_{2c} := \frac{N_{c2}}{l_{c2} \cdot b_{c2}} - \frac{6 \cdot M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 156.588\ \frac{kN}{m^2}$$

$a := 0.02\ m$ $\quad c := \frac{l_{c2}}{2} - \frac{M_{xc2}}{N_{c2}} = 0.618\ m$

$a \le 0.3 \cdot l_{c2} = 1$

$$P_{max} := \frac{2 \cdot N_{c2}}{3 \cdot b_{c2} \cdot c} = 218.462\ kPa$$

$$T := \frac{1}{2} \cdot \left(-P_{2c}\right) \cdot a \cdot b_{c2} = -1.566 \ kN$$

$$f_{yd} := 500 \ MPa \frac{}{1.15} = 434.783 \ MPa$$

$$A_{sl.nec} \geq \frac{T}{f_{yd}}$$

$$A_{sl.nec} := \frac{T}{f_{yd}} = -0.036 \ cm^2$$

$$A_{s.eff} := 3.14 \ cm^2$$ **ALEG 4fi10/220**

$$P_{lc1} := 1782.963 \ kPa$$

$$l_{c1} := 0.15 \ m$$

$$M_1 := \left(P_{lc1} \cdot l_{c1} \cdot \frac{l_{c1}}{2} + \frac{\left(P_{max} - P_{lc1}\right) \cdot l_{c1}}{2} \cdot \frac{2}{3} \cdot l_{c1}\right) \cdot b_{c2} = 8.325 \ kN \cdot m$$

$$P_{med} := \frac{P_{max}}{2} = 109.231 \ kPa$$

$$l_{c22} := 0.125 \ m$$

$$M_2 := \left(P_{med} \cdot l_{c22} \cdot \frac{l_{c22}}{2}\right) \cdot l_{c2} = 1.067 \ kN \cdot m$$

MARCA 1 **CLASĂ BETON C16/20** **OȚEL S500**

$f_{ck} := 16 \ MPa$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$$f_{cd} := \frac{f_{ck}}{1.5} = 10.667 \ MPa$$

$$f_{yk} := 500 \ MPa$$

$$f_{yd} := \frac{f_{yk}}{1.15} = 434.783 \ MPa$$

$M_{ed} := M_1$ $d := h_{c2} - 2 \cdot c_{nom} = 31 \ cm$

$$\mu := \frac{M_{ed}}{l_{c2} \cdot d^2 \cdot f_{cd}} = 0.006 \quad => \quad \mu \le \mu_{lim}(0.372)$$

$$w := 1 - \sqrt{1 - 2 \cdot \mu} = 0.007$$

$$A_{nec} := w \cdot l_{c2} \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.62 \ cm^2$$

$$A_{smin} := 0.001 \cdot h_{c2} \cdot b_{c2} = 4 \ cm^2$$

$$A_{seff1} := 3.14 \ cm^2$$ **ALEG 4fi10/160**

MARCA 2 **CLASĂ BETON C16/20** **OȚEL S500**

$M_{ed2} := M_2$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$$h_{c2} := 0.3 \ m$$

$$\mu_2 := \frac{M_{ed2}}{b_{c2} \cdot d^2 \cdot f_{cd}} = 0.001 \quad => \quad \mu \le \mu_{lim}(0.372)$$

$$w_2 := 1 - \sqrt{1 - 2 \cdot \mu_2} = 0.001$$

$$A_{nec2} := w_2 \cdot b_{c2} \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.079 \ cm^2$$

$$A_{smin2} := 0.001 \cdot h_{c2} \cdot l_{c2} = 3.75 \ cm^2$$

$$A_{seff2} := 3.14 \ cm^2$$ **ALEG 4fi10/220**

PREDIMENSIONARE STÂLP S3 (stâlp central)

$A_{s3} := 0.5\ m \cdot 0.4\ m = 0.2\ m^2$

$N_{s3} := A_{s3} \cdot \left(P_d + Q_d\right) = 6.426\ kN$

$M_{x3} := 0.2\ m \cdot N_{s3} = 1.285\ kN \cdot m$

$M_{y3} := 0.1\ m \cdot N_{s3} = 0.643\ kN \cdot m$

$T_{y3} := 0.05 \cdot N_{s3} = 0.321\ kN$

$T_{x3} := 0.1 \cdot N_{s3} = 0.643\ kN$

$H_i := 0.9\ m$

$D_{f3} := H_i + 20\ cm = 1.1\ m$

Dimensiunea tălpii fundației: L și B

$P_{eff} \le P_{adm}$

$P_{conv} := 297.25\ kPa \qquad P_{adm} := 1.2\ P_{conv} = 356.7\ kPa$

$V_{d3} := 1.2 \cdot N_{s3} = 7.711\ kN$

$r_3 := \frac{l_{st}}{b_{st}} = 1.25$ =L/B=>L=B*r3

$\frac{\left(1.2 \cdot N_{s3}\right)}{B_3 \cdot L_3} \le P_{adm}$=> $\frac{\left(1.2 \cdot N_{s3}\right)}{B_3 \cdot L_3} \le P_{adm}$=> $B_{3min} := \sqrt{\frac{V_{d3}}{r_3 \cdot P_{adm}}} = 0.132\ m$

$B_3 := 2.00\ m \qquad L_3 := r_3 \cdot B_3 = 2.5\ m$

Înălțimea blocului de fundare și a cuzinetului: H și hc

$\frac{H_3}{L_3} \ge 0.3 \qquad H_3 := L_3 \cdot 0.3 = 0.75\ m \qquad$ => $\ alegem\ H_3 := 0.85\ m$

$hc3 := 0.40\ m$

VERIFICAREA LA CAPACITATE PORTANTĂ A TĂLPII FUNDAȚIEI

$P_{eff} \leq P_{adm}$

$\frac{V_d}{A'} \leq \frac{R_d}{A'}$

Efortul vertical de calcul la talpa fundației

$L_{af3} := 6\ m$

$\gamma_{BCA} := 16\ \frac{kN}{m^3}$ $\gamma_{ba} := 24\ \frac{kN}{m^3}$ $\gamma_{med} := 22\ \frac{kN}{m^3}$

$H_{tot3} := D_{f3} = 1.1\ m$

$N_{ed3} := N_{s3} = 6.426\ kN$

$G_{f3} := B_3 \cdot L_3 \cdot H_{tot3} \cdot \gamma_{med} \cdot \gamma_G = 163.35\ kN$

$V_{d3} := N_{ed3} + G_{f3} = 169.776\ kN$

Momentele încovoietoare la talpa fundației

$M_{xf3} := M_{x3} + T_{y3} \cdot H_3 = 1.558\ kN \cdot m$

$M_{yf3} := M_{y3} + T_{x3} \cdot H_3 = 1.189\ kN \cdot m$

Forțele orizontale de la talpa fundației

$T_{xf3} := T_{x3} = 0.643\ kN$

$T_{yf3} := T_{y3} = 0.321\ kN$

$H_{d3} := \sqrt{T_{xf3}^{\ 2} + T_{yf3}^{\ 2}} = 0.718\ kN$

Calculul presiunii efective la talpa fundației

$e_{B3} := \frac{M_{yf3}}{V_{d3}} = 0.007\ m$

$e_{L3} := \frac{M_{xf3}}{V_{d3}} = 0.009\ m$

VERIFICAREA EXCENTRICITĂȚII

$$\left(\frac{e_{B3}}{B_3}\right)^2+\left(\frac{e_{L3}}{L_3}\right)^2\le\frac{1}{9}=1$$ **VERIFICĂ**

$B'_3:=B_3-2\cdot e_{B3}=1.986\ m$

$L'_3:=L_3-2\cdot e_{L3}=2.482\ m$

$A'_3:=B'_3\cdot L'_3=4.929\ m^2$

$\gamma_{k2}:=18.7\ \frac{kN}{m^3}$ $c'_{k2}:=29.4\ kPa$ $\gamma_{c'}:=1.25$ $\gamma_\gamma:=1$

$c'_{d2}:=\frac{c'_{k2}}{\gamma_{c'}}=23.52\ \frac{kN}{m^2}$ *coeziunea terenului de sub talpa fundației*

$\gamma':=\frac{\gamma_{k2}}{\gamma_\gamma}=18.7\ \frac{kN}{m^3}$ *greutatea volumica a pamântului de sub talpa fundației*

$q':=D_{f3}\cdot\gamma'=20.57\ \frac{kN}{m^2}$ *suprasarcina de calcul care acționează la talpa fundației*

$\phi'_{k2}:=15°$ $\tan\left(\phi'_{k2}\right)=0.268$

$N_q:=e^{\pi\cdot\tan\left(\phi'_{k2}\right)}\cdot\left(\tan\left(45°+\frac{\phi'_{k2}}{2}\right)\right)^2=3.941$

$N_c:=\left(N_q-1\right)\cdot\frac{1}{\tan\left(\phi'_{k2}\right)}=10.977$ $m_B:=\frac{2+\frac{B'_3}{L'_3}}{1+\frac{B'_3}{L'_3}}=1.555$

$N_\gamma:=2\cdot\left(N_q-1\right)\cdot\tan\left(\phi'_{k2}\right)=1.576$ $m_L:=\frac{2+\frac{L'_3}{B'_3}}{1+\frac{L'_3}{B'_3}}=1.445$

$b_q:=1$ $b_\gamma:=1$

$b_c:=b_q-\frac{\left(1-b_q\right)}{\left(N_c\cdot\tan\left(\phi'_{k2}\right)\right)}=1$

$$s_q := 1 + \left(\frac{B'_3}{L'_3}\right) \cdot \sin\left(\phi'_{k2}\right) = 1.207$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_3}{L'_3}\right) = 0.76 \qquad \theta := \operatorname{atan}\left(\frac{T_{y3}}{T_{x3}}\right) = 26.565\ \textit{deg}$$

$$s_c := \frac{\left(s_q \cdot N_q - 1\right)}{\left(N_q - 1\right)} = 1.278 \qquad m_\theta := m_L \cdot \cos\left(\theta\right)^2 + m_B \cdot \sin\left(\theta\right)^2 = 1.467$$

$$i_q := \left(1 - \frac{0.5}{\left(550 + 1.338 \cdot 23.52 \cdot \cos\left(\phi'_{k2}\right)\right)}\right)^{m_\theta} = 0.999$$

$$i_\gamma := \left(1 - \frac{0.5}{\left(550 + 1.338 \cdot 29.4 \cdot \cos\left(\phi'_{k2}\right)\right)}\right)^{m_\theta} = 0.999$$

$$i_c := i_q - \frac{\left(1 - i_q\right)}{N_c \cdot \tan\left(\phi'_{k2}\right)} = 0.998 \qquad P_{adm3} := \frac{R_d}{A'_3}$$

$$P_{adm3} := c'_{d2} \cdot N_c \cdot b_c \cdot s_c \cdot i_c + q' \cdot N_q \cdot b_q \cdot s_q \cdot i_q + 0.5 \cdot \gamma' \cdot B'_3 \cdot N_\gamma \cdot b_\gamma \cdot s_\gamma \cdot i_\gamma = 449.214\ \frac{\textit{kN}}{\textit{m}^2}$$

$$\frac{V_{d3}}{A'_3} = 34.448\ \frac{\textit{kN}}{\textit{m}^2} \qquad P_{adm3} > \frac{V_{d3}}{A'_3}$$

VERIFICĂ

VERIFICAREA STRUCTURALĂ A FIE - ARMAREA TĂLPII

$L_{af3} := 6\ \textit{m}$

$H_{tot3} := 1.25\ \textit{m}$

$B_3 = 2\ \textit{m}$

$L_3 = 2.5\ \textit{m}$

$\gamma_{BCA} := 16\ \frac{\textit{kN}}{\textit{m}^3}$

$\gamma_{b.a} := 24\ \frac{\textit{kN}}{\textit{m}^3}$

$\gamma_{med} := 22\ \frac{\textit{kN}}{\textit{m}^3}$

$N_{s3} = 6.426\ \textit{kN}$

$M_{xaf3} := M_{x3} + T_{y3} \cdot H_3 = 1.558\ \textit{kN} \cdot \textit{m}$

$M_{yaf3} := M_{y2} + T_{x2} \cdot H_3 = 1.189\ \textit{kN} \cdot \textit{m}$

$G_{f3} := 77.517\ kN$

$V_{d3} := N_{s3} + G_{f3} = 83.943\ kN$

$$P_{13} := \frac{V_{d3}}{L_3 \cdot B_3} + \frac{6 \cdot M_{xaf3}}{{L_3}^2 \cdot B_3} = 17.537\ \frac{kN}{m^2}$$

$$P_{23} := \frac{V_{d3}}{L_3 \cdot B_3} - \frac{6 \cdot M_{xaf3}}{{L_3}^2 \cdot B_3} = 16.041\ \frac{kN}{m^2}$$

$P_{w1} := P_{13}$ $w_1 := 65\ cm$ $w_2 := 52.5\ cm$

$$M_{13} := \left(\left(P_{w1} \cdot w_1 \cdot \frac{w_1}{2}\right) + \frac{\left(P_{13} - P_{w1}\right) \cdot w_1}{2} \cdot \frac{2}{3} \cdot w_1\right) \cdot B_3 = 7.409\ kN \cdot m$$

$$P_{med3} := \frac{\left(P_{13} + P_{23}\right)}{2} = 16.789\ \frac{kN}{m^2}$$

$$M_{23} := \left(P_{med3} \cdot w_2 \cdot \frac{w_2}{2}\right) \cdot L_2 = 5.784\ kN \cdot m$$

MARCA 3 **CLASĂ BETON C16/20** **OȚEL S500**

$f_{ck} := 16\ MPa$ **Strat acoperire beton** $c_{nom} := 4.5\ cm$

$$f_{cd} := \frac{f_{ck}}{1.5} = 10.667\ MPa$$

$f_{yk} := 500\ MPa$

$$f_{yd} := \frac{f_{yk}}{1.15} = 434.783\ MPa$$

$M_{ed} := M_{13}$

$d_3 := c_{nom} + 1\ cm = 0.055\ m$

$d := H_3 - c_{nom} + 10\ cm = 0.905\ m$

$$\mu := \frac{M_{ed}}{L_3 \cdot d^2 \cdot f_{cd}} = 3.392 \cdot 10^{-4} \Rightarrow \quad \mu \le \mu_{lim}\ (0.372)$$

$$w := 1 - \sqrt{1 - 2 \cdot \mu} = 3.393 \cdot 10^{-4}$$

$$A_{nec} := w \cdot L_3 \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.188 \ cm^2$$

$$A_{s.min} := 0.001 \cdot H_3 \cdot B_3 = 17 \ cm^2$$

$A_{seff3} := 8.635 \ cm^2$ **ALEG 11fi10/150**

MARCA 4 **CLASĂ BETON C16/20** **OȚEL S500**

$M_{ed3} := M_{23}$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$H_3 := 0.55 \ m$

$$\mu := \frac{M_{ed3}}{B_3 \cdot d^2 \cdot f_{cd}} = 3.31 \cdot 10^{-4} \quad => \quad \mu \leq \mu_{lim}(0.372)$$

$$w_3 := 1 - \sqrt{1 - 2 \cdot \mu} = 3.311 \cdot 10^{-4}$$

$$A_{nec} := w_3 \cdot B_3 \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.147 \ cm^2$$

$$A_{s.min} := 0.001 \cdot h_{c2} \cdot L_3 = 7.5 \ cm^2$$

$A_{seff2} := 5.49 \ cm^2$ **ALEG 7fi10/240**

VERIFICARE LA TASARE

$S_{Ed} \leq S_{Cd}$ $s_{adm} := 80 \ mm$

$S_{Cd} := s_{adm}$

$$P_{net2} := \frac{V_{d2}}{A'_2} - D_{f2} \cdot \gamma_{k2} = 78.595 \ kPa$$

$$P_{net3} := \frac{V_{d3}}{A'_3} - D_{f3} \cdot \gamma_{k2} = -3.538 \ kPa$$

$S_{Ed2} := 0.0346563786 \ m = 34.656 \ mm$ $S_{Ed2} < S_{Cd} = 1$

$S_{Ed3} := 0.03645053672 \ m = 36.451 \ mm$ $S_{Ed3} < S_{Cd} = 1$

VERIFICĂ

TEMA 3
FUNDAȚII IZOLATE

DATE PROIECT

$n := 4.5$
$a := 15\ cm$
$H_z := 3\ m$
$C.T.S := -(0.55\ m + a) = -0.7\ m$

Stratificația terenului

Umplutură

$h_1 := 0.5\ m$

$\gamma_{k1} := 17.2\ \frac{kN}{m^3}$

Argilă prăfoasă cafenie, plastic consistentă

$h_2 := 3\ m$

$\gamma_{k2} := 18.7\ \frac{kN}{m^3}$

$w_2 := 22\%$

$I_{p2} := 22\%$

$I_{c2} := 0.70$

$\phi'_{k2} := 15\ deg$

$e_2 := 0.63$

$E_2 := 28.000\ kPa$

$c'_{k2} := 29.45\ kPa$

Nisip argilos galben cafeniu

$h_3 := 6.5\ m$

$\gamma_{k3} := 19.3\ \frac{kN}{m^3}$

$w := 18\%$

$I_{p3} := 11\%$

$\phi'_{k3} := 31\ deg$

$e_3 := 0.6$

$E_3 := 42.000\ kPa$

$c'_{k3} := 9.25\ kPa$

Încărcări de calcul pentru fiecare stâlp

- încărcarea distribuită permanentă: $P_k := 16.4\ kPa$

- încărcarea distribuită variabilă: $Q_k := 7.4\ kPa$

NOTE DE CALCUL

$\gamma_G := 1.35$ $\quad P_d := P_k \cdot \gamma_G = 22.14\ kPa$ $\quad H := 3\ m$ $\quad l_{st} := 0.5\ m$

$\gamma_Q := 1.35$ $\quad Q_d := Q_k \cdot \gamma_Q = 9.99\ kPa$ $\quad b_z := 0.3\ m$ $\quad b_{st} := 0.4\ m$

PREDIMENSIONARE STÂLP S1 (stâlp de colț) + S2 (stâlp marginal)

Eforturi secționale

$A_{S2} := 0.5\ m \cdot 0.4\ m = 0.2\ m^2$

$N_{S2} := A_{S2} \cdot \left(P_d + Q_d\right) = 6.426\ kN$

$M_{x2} := 0.2\ m \cdot N_{S2} = 1.285\ kN \cdot m$

$M_{y2} := 0.1\ m \cdot N_{S2} = 0.643\ kN \cdot m$

$T_{x2} := 0.1 \cdot N_{S2} = 0.643\ kN$

$T_{y2} := 0.05 \cdot N_{S2} = 0.321\ kN$

$H_i := 0.90\ m$ $\quad D_{f2} := H_i + 20\ cm = 1.1\ m$

Dimensiunea tălpii fundației: L și B

$P_{eff} \le P_{adm}$

$P_{conv} := 297.25\ kPa$ $\quad P_{adm} := 1.2\ P_{conv} = 356.7\ kPa$

$V_{d2} := 1.2\ N_{S2} = 7.711\ kN$

$r_2 := \dfrac{l_{st}}{b_{st}} = 1.25$

$\dfrac{1.2 \cdot N}{B_2 \cdot L_2} \le P_{adm}$ $\quad \dfrac{1.2\ N_{S2}}{{B_2}^2 \cdot r_2} \le P_{adm}$ $\quad B_{2min} := \sqrt{\left(\dfrac{V_{d2}}{r_2 \cdot P_{adm}}\right)} = 0.132\ m$

$B_2 := 2\ m$

$L_2 := r_2 \cdot B_2 = 2.5\ m$

Dimensiunea tălpii cuzinetului: lc și bc

$l_{c2min} := 0.5 \cdot L_2 = 1.25\ m$ alegem $l_{c2} := 1.25\ m$

$b_{c2min} := 0.5 \cdot B_2 = 1\ m$ alegem $b_{c2} := 1.00\ m$

Înălțimea blocului de fundare și a cuzinetului: H și hc

$tan\alpha \geq tan\alpha_{adm}$ $tan\alpha_{adm} := 1.3$

$H_2 := tan\alpha_{adm} \cdot \frac{(L_2 - l_{c2})}{2} = 0.813\ m$ $tan\alpha := \frac{H_2}{\frac{L_2 - l_{c2}}{2}} = 1.3$

$H_2 := tan\alpha_{adm} \cdot \frac{(B_2 - b_{c2})}{2} = 0.65\ m$ $tan\alpha := \frac{H_2}{\frac{B_2 - b_{c2}}{2}} = 1.3$

alegem $H_2 := 0.85\ m$

$tan\beta \geq tan\beta_{adm}$

$tan\beta_{adm} := 1$ $tan\beta := tan\beta_{adm}$

$h_{c2} := tan\beta \cdot \frac{(l_{c2} - l_{st})}{2} = 0.375\ m$ $tan\beta_2 := \frac{l_{c2}}{\frac{(l_{c2} - l_{st})}{2}} = 3.333$

$h_{c2} := tan\beta \cdot \frac{(b_{c2} - b_{st})}{2} = 0.3\ m$ $tan\beta_2 := \frac{b_{c2}}{\frac{(b_{c2} - b_{st})}{2}} = 3.333$

alegem $h_{c2} := 0.40\ m$

$\frac{l_{c2}}{h_{c2}} \geq 2.5$ **VERIFICĂ**

VERIFICAREA LA CAPACITATE PORTANTĂ A TĂLPII FUNDAȚIEI

$P_{eff} \leq P_{adm}$

$\frac{V_d}{A'} \leq \frac{R_d}{A'}$

Efortul vertical de calcul la talpa fundației

$L_{af2} := 6\ m$

$\gamma_{BCA} := 16\ \frac{kN}{m^3}$ $\gamma_{b.a} := 24\ \frac{kN}{m^3}$ $\gamma_{med} := 22\ \frac{kN}{m^3}$

$b_{gr} := b_z = 0.3\ m$

$1.5 \leq \frac{h_{gr}}{b_{gr}} \leq 3$ $h_{gr} := 0.85\ m$ $\frac{h_{gr}}{b_{gr}} = 2.833$ **VERIFICĂ**

$H_{tot} := H_2 + h_{c2} + h_{gr} = 2.1\ m$

$N_{cd2} := N_{S2} = 6.426\ kN$

$G_z := b_z \cdot H \cdot L_{af2} \cdot \gamma_{BCA} \cdot \gamma_G = 116.64\ kN$

$G_{f2} := B_2 \cdot L_2 \cdot H_{tot} \cdot \gamma_{med} \cdot \gamma_G = 311.85\ kN$

$G_{gr} := b_{gr} \cdot h_{gr} \cdot L_{af2} \cdot \gamma_{b.a} \cdot \gamma_G = 49.572\ kN$

$V_{d2} := N_{cd2} + G_{f2} + G_{gr} + G_z = 484.488\ kN$

Momentele încovoietoare la talpa fundației

$M_{xf2} := M_{x2} + T_{y2} \cdot (h_{c2} + H_2) = 1.687\ kN \cdot m$

$M_{yf2} := M_{y2} + T_{x2} \cdot (h_{c2} + H_2) + (G_z + G_{gr}) \cdot \left(\frac{b_{st}}{2} - \frac{b_z}{2}\right) = 9.756\ kN \cdot m$

Forțele orizontale de la talpa fundației

$T_{xf2} := T_{x2} = 0.643\ kN$

$T_{yf2} := T_{y2} = 0.321\ kN$

$H_{d2} := \sqrt{T_{xf2}{}^2 + T_{yf2}{}^2} = 0.718\ kN$

Calculul presiunii efective la talpa fundației

$e_B := \frac{M_{yf2}}{V_{d2}} = 0.02\ m$

$e_L := \frac{M_{xf2}}{V_{d2}} = 0.003\ m$

VERIFICAREA EXCENTRICITĂȚII

$\left(\frac{e_B}{B_2}\right)^2 + \left(\frac{e_L}{L_2}\right) \le \frac{1}{9} = 1$ **VERIFICĂ**

$B'_2 := B_2 - 2 \cdot e_B = 1.96\ m$

$L'_2 := L_2 - 2\ e_L = 2.493\ m$

$A'_2 := B'_2 \cdot L'_2 = 4.886\ m^2$

$\gamma_{c'} := 1.25 \qquad \gamma_\gamma := 1$

$c'_{d2} := \frac{c'_{k2}}{\gamma_{c'}} = 23.56\ \frac{kN}{m^2}$

$\gamma' := \frac{\gamma_{k2}}{\gamma_\gamma} = 18.7\ \frac{kN}{m^3}$

$q' := D_{f2} \cdot \gamma' = 20.57\ \frac{kN}{m^2}$

$\tan\left(\phi'_{k2}\right) = 0.268$

$N_q := e^{\pi \cdot \tan\left(\phi'_{k2}\right)} \cdot \left(\tan\left(45\ ° + \frac{\phi'_{k2}}{2}\right)\right)^2 = 3.941 \qquad N_q := 3.941$

$$N_c := (N_q - 1) \cdot \frac{1}{\tan(\phi'_{k2})} = 10.976 \qquad m_B := \frac{2 + \frac{B'_2}{L'_2}}{1 + \frac{B'_2}{L'_2}} = 1.56$$

$$N_\gamma := 2\ (N_q - 1) \cdot \tan(\phi'_{k2}) = 1.576 \qquad m_L := \frac{2 + \frac{L'_2}{B'_2}}{1 + \frac{L'_2}{B'_2}} = 1.44$$

$$b_q := 1 \qquad b_\gamma := 1$$

$$b_c := b_q - \frac{(1 - b_q)}{N_c \cdot \tan(\phi'_{k2})} = 1 \qquad \theta := \operatorname{atan}\left(\frac{T_{y2}}{T_{x2}}\right) = 26.565\ ^\circ$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_2}{L'_2}\right) = 0.764$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_2}{L'_2}\right) = 0.764$$

$$s_q := 1 + \frac{B'_2}{L'_2} \cdot \sin(\phi'_{k2}) = 1.203 \qquad m_\theta := m_L \cdot \left(\cos(\theta)^2 + m_B \cdot \sin(\theta)^2\right) = 1.601$$

$$s_c := \frac{s_q \cdot N_q - 1}{N_q - 1} = 1.273$$

$$i_q := \left(1 - \left(\frac{0.5}{550 + 1.338 \cdot 23.52 \cdot \cos(\phi'_{k2})}\right)\right)^{m_\theta} = 0.999$$

$$i_\gamma := \left(1 - \left(\frac{0.5}{550 + 1.338 \cdot 29.4 \cdot \cos(\phi'_{k2})}\right)\right)^{m_\theta} = 0.999$$

$$i_c := i_q - \frac{(1 - i_q)}{N_c \cdot \tan(\phi'_{k2})} = 0.998 \qquad P_{adm2} := \frac{R_d}{A'_2}$$

$$P_{adm2} := c'_{d2} \cdot N_c \cdot b_c \cdot s_c \cdot i_c + q' \cdot N_q \cdot b_q \cdot s_q \cdot i_q + 0.5 \cdot \gamma' \cdot B'_2 \cdot N_\gamma \cdot b_\gamma \cdot s_\gamma \cdot i_\gamma = 447.95\ \frac{kN}{m^2}$$

$$\frac{V_{d2}}{A'_2} = 99.165\ \frac{kN}{m^2} \qquad P_{adm2} > \frac{V_{d2}}{A'_2}$$

VERIFICĂ

ARMAREA CUZINETULUI

$b_{c2} := 1.00\ m$ $\quad L_{af2} = 6\ m$ $\quad \gamma_{bca} := 16\ \frac{kN}{m^3}$

$l_{c2} := 1.25\ m$ $\quad h_{gr} = 0.85\ m$ $\quad \gamma b.a. := 24\ \frac{kN}{m^3}$

$h_{c2} := 0.4\ m$ $\quad b_{gr} = 0.3\ m$ $\quad \gamma_{med} := 22\ \frac{kN}{m^3}$

$H_{tot.c2} := h_{gr} + h_{c2} = 1.25\ m$

$M_{xc2} := M_{x2} + T_{y2} \cdot h_{c2} = 1.414\ kN \cdot m$

$M_{yc2} := M_{y2} + T_{x2} \cdot h_{c2} = 0.9\ kN \cdot m$

$G_{gr} := b_{gr} \cdot h_{gr} \cdot L_{af2} \cdot \gamma_{bca} \cdot \gamma_G = 33.048\ kN$

$G_z := b_z \cdot H \cdot L_{af2} \cdot \gamma_{bca} \cdot \gamma_G = 116.64\ kN$

$G_{c2} := H_{tot.c2} \cdot b_{c2} \cdot l_{c2} \cdot \gamma_{med} \cdot \gamma_G = 46.406\ kN$

$N_{c2} := N_{S2} + G_{gr} + G_z + G_{c2} = 202.52\ kN$

$A_{c2} := b_{c2} \cdot l_{c2} = 1.25\ m^2$

$$P_{1c} := \frac{N_{c2}}{b_{c2} \cdot l_{c2}} + 6 \cdot \frac{M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 167.445\ \frac{kN}{m^2}$$

$$P_{2c} := \frac{N_{c2}}{l_{c2} \cdot b_{c2}} - \frac{6 \cdot M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 156.588\ \frac{kN}{m^2}$$

$$P_{2c} := \frac{N_{c2}}{l_{c2} \cdot b_{c2}} - \frac{6 \cdot M_{xc2}}{{l_{c2}}^2 \cdot b_{c2}} = 156.588\ \frac{kN}{m^2}$$

$a := 0.02\ m$ $\quad c := \frac{l_{c2}}{2} - \frac{M_{xc2}}{N_{c2}} = 0.618\ m$

$a \le 0.3 \cdot l_{c2} = 1$

$$P_{max} := \frac{2 \cdot N_{c2}}{3 \cdot b_{c2} \cdot c} = 218.462\ kPa$$

$$T := \frac{1}{2} \cdot \left(-P_{2c}\right) \cdot a \cdot b_{c2} = -1.566 \ kN$$

$$f_{yd} := 500 \ MPa \frac{}{1.15} = 434.783 \ MPa$$

$$A_{sl.nec} \geq \frac{T}{f_{yd}}$$

$$A_{sl.nec} := \frac{T}{f_{yd}} = -0.036 \ cm^2$$

$$A_{s.eff} := 3.14 \ cm^2$$ **ALEG 4fi10/220**

$$P_{lc1} := 1782.963 \ kPa$$

$$l_{c1} := 0.15 \ m$$

$$M_1 := \left(P_{lc1} \cdot l_{c1} \cdot \frac{l_{c1}}{2} + \frac{\left(P_{max} - P_{lc1}\right) \cdot l_{c1}}{2} \cdot \frac{2}{3} \cdot l_{c1}\right) \cdot b_{c2} = 8.325 \ kN \cdot m$$

$$P_{med} := \frac{P_{max}}{2} = 109.231 \ kPa$$

$$l_{c22} := 0.125 \ m$$

$$M_2 := \left(P_{med} \cdot l_{c22} \cdot \frac{l_{c22}}{2}\right) \cdot l_{c2} = 1.067 \ kN \cdot m$$

MARCA 1 **CLASĂ BETON C16/20** **OȚEL S500**

$f_{ck} := 16 \ MPa$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$$f_{cd} := \frac{f_{ck}}{1.5} = 10.667 \ MPa$$

$$f_{yk} := 500 \ MPa$$

$$f_{yd} := \frac{f_{yk}}{1.15} = 434.783 \ MPa$$

$M_{ed} := M_1$ $d := h_{c2} - 2 \cdot c_{nom} = 31 \ cm$

$$\mu := \frac{M_{ed}}{l_{c2} \cdot d^2 \cdot f_{cd}} = 0.006 \quad => \quad \mu \le \mu_{lim}(0.372)$$

$$w := 1 - \sqrt{1 - 2 \cdot \mu} = 0.007$$

$$A_{nec} := w \cdot l_{c2} \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.62 \ cm^2$$

$$A_{smin} := 0.001 \cdot h_{c2} \cdot b_{c2} = 4 \ cm^2$$

$$A_{seff1} := 3.14 \ cm^2$$ **ALEG 4fi10/160**

MARCA 2 **CLASĂ BETON C16/20** **OȚEL S500**

$M_{ed2} := M_2$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$$h_{c2} := 0.3 \ m$$

$$\mu_2 := \frac{M_{ed2}}{b_{c2} \cdot d^2 \cdot f_{cd}} = 0.001 \quad => \quad \mu \le \mu_{lim}(0.372)$$

$$w_2 := 1 - \sqrt{1 - 2 \cdot \mu_2} = 0.001$$

$$A_{nec2} := w_2 \cdot b_{c2} \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.079 \ cm^2$$

$$A_{smin2} := 0.001 \cdot h_{c2} \cdot l_{c2} = 3.75 \ cm^2$$

$$A_{seff2} := 3.14 \ cm^2$$ **ALEG 4fi10/220**

PREDIMENSIONARE STÂLP S3 (stâlp central)

$A_{s3} := 0.5\ m \cdot 0.4\ m = 0.2\ m^2$

$N_{s3} := A_{s3} \cdot \left(P_d + Q_d\right) = 6.426\ kN$

$M_{x3} := 0.2\ m \cdot N_{s3} = 1.285\ kN \cdot m$

$M_{y3} := 0.1\ m \cdot N_{s3} = 0.643\ kN \cdot m$

$T_{y3} := 0.05 \cdot N_{s3} = 0.321\ kN$

$T_{x3} := 0.1 \cdot N_{s3} = 0.643\ kN$

$H_i := 0.9\ m$

$D_{f3} := H_i + 20\ cm = 1.1\ m$

Dimensiunea tălpii fundației: L și B

$P_{eff} \le P_{adm}$

$P_{conv} := 297.25\ kPa \qquad P_{adm} := 1.2\ P_{conv} = 356.7\ kPa$

$V_{d3} := 1.2 \cdot N_{s3} = 7.711\ kN$

$r_3 := \frac{l_{st}}{b_{st}} = 1.25$ =L/B=>L=B*r3

$\frac{\left(1.2 \cdot N_{s3}\right)}{B_3 \cdot L_3} \le P_{adm}$=> $\frac{\left(1.2 \cdot N_{s3}\right)}{B_3 \cdot L_3} \le P_{adm}$=> $B_{3min} := \sqrt{\frac{V_{d3}}{r_3 \cdot P_{adm}}} = 0.132\ m$

$B_3 := 2.00\ m \qquad L_3 := r_3 \cdot B_3 = 2.5\ m$

Înălțimea blocului de fundare și a cuzinetului: H și hc

$\frac{H_3}{L_3} \ge 0.3 \qquad H_3 := L_3 \cdot 0.3 = 0.75\ m \qquad$ => $\ alegem\ H_3 := 0.85\ m$

$hc3 := 0.40\ m$

VERIFICAREA LA CAPACITATE PORTANTĂ A TĂLPII FUNDAȚIEI

$P_{eff} \leq P_{adm}$

$\frac{V_d}{A'} \leq \frac{R_d}{A'}$

Efortul vertical de calcul la talpa fundației

$L_{af3} := 6\ m$

$\gamma_{BCA} := 16\ \frac{kN}{m^3}$ $\gamma_{ba} := 24\ \frac{kN}{m^3}$ $\gamma_{med} := 22\ \frac{kN}{m^3}$

$H_{tot3} := D_{f3} = 1.1\ m$

$N_{ed3} := N_{s3} = 6.426\ kN$

$G_{f3} := B_3 \cdot L_3 \cdot H_{tot3} \cdot \gamma_{med} \cdot \gamma_G = 163.35\ kN$

$V_{d3} := N_{ed3} + G_{f3} = 169.776\ kN$

Momentele încovoietoare la talpa fundației

$M_{xf3} := M_{x3} + T_{y3} \cdot H_3 = 1.558\ kN \cdot m$

$M_{yf3} := M_{y3} + T_{x3} \cdot H_3 = 1.189\ kN \cdot m$

Forțele orizontale de la talpa fundației

$T_{xf3} := T_{x3} = 0.643\ kN$

$T_{yf3} := T_{y3} = 0.321\ kN$

$H_{d3} := \sqrt{T_{xf3}^{\ 2} + T_{yf3}^{\ 2}} = 0.718\ kN$

Calculul presiunii efective la talpa fundației

$e_{B3} := \frac{M_{yf3}}{V_{d3}} = 0.007\ m$

$e_{L3} := \frac{M_{xf3}}{V_{d3}} = 0.009\ m$

VERIFICAREA EXCENTRICITĂȚII

$$\left(\frac{e_{B3}}{B_3}\right)^2+\left(\frac{e_{L3}}{L_3}\right)^2\le\frac{1}{9}=1$$ **VERIFICĂ**

$B'_3:=B_3-2\cdot e_{B3}=1.986\ m$

$L'_3:=L_3-2\cdot e_{L3}=2.482\ m$

$A'_3:=B'_3\cdot L'_3=4.929\ m^2$

$\gamma_{k2}:=18.7\ \frac{kN}{m^3}$ $c'_{k2}:=29.4\ kPa$ $\gamma_{c'}:=1.25$ $\gamma_\gamma:=1$

$c'_{d2}:=\frac{c'_{k2}}{\gamma_{c'}}=23.52\ \frac{kN}{m^2}$ *coeziunea terenului de sub talpa fundației*

$\gamma':=\frac{\gamma_{k2}}{\gamma_\gamma}=18.7\ \frac{kN}{m^3}$ *greutatea volumica a pamântului de sub talpa fundației*

$q':=D_{f3}\cdot\gamma'=20.57\ \frac{kN}{m^2}$ *suprasarcina de calcul care acționează la talpa fundației*

$\phi'_{k2}:=15°$ $\tan\left(\phi'_{k2}\right)=0.268$

$N_q:=e^{\pi\cdot\tan\left(\phi'_{k2}\right)}\cdot\left(\tan\left(45°+\frac{\phi'_{k2}}{2}\right)\right)^2=3.941$

$N_c:=\left(N_q-1\right)\cdot\frac{1}{\tan\left(\phi'_{k2}\right)}=10.977$ $m_B:=\frac{2+\frac{B'_3}{L'_3}}{1+\frac{B'_3}{L'_3}}=1.555$

$N_\gamma:=2\cdot\left(N_q-1\right)\cdot\tan\left(\phi'_{k2}\right)=1.576$ $m_L:=\frac{2+\frac{L'_3}{B'_3}}{1+\frac{L'_3}{B'_3}}=1.445$

$b_q:=1$ $b_\gamma:=1$

$b_c:=b_q-\frac{\left(1-b_q\right)}{\left(N_c\cdot\tan\left(\phi'_{k2}\right)\right)}=1$

$$s_q := 1 + \left(\frac{B'_3}{L'_3}\right) \cdot \sin\left(\phi'_{k2}\right) = 1.207$$

$$s_\gamma := 1 - 0.3 \cdot \left(\frac{B'_3}{L'_3}\right) = 0.76 \qquad \theta := \operatorname{atan}\left(\frac{T_{y3}}{T_{x3}}\right) = 26.565\ \textit{deg}$$

$$s_c := \frac{\left(s_q \cdot N_q - 1\right)}{\left(N_q - 1\right)} = 1.278 \qquad m_\theta := m_L \cdot \cos\left(\theta\right)^2 + m_B \cdot \sin\left(\theta\right)^2 = 1.467$$

$$i_q := \left(1 - \frac{0.5}{\left(550 + 1.338 \cdot 23.52 \cdot \cos\left(\phi'_{k2}\right)\right)}\right)^{m_\theta} = 0.999$$

$$i_\gamma := \left(1 - \frac{0.5}{\left(550 + 1.338 \cdot 29.4 \cdot \cos\left(\phi'_{k2}\right)\right)}\right)^{m_\theta} = 0.999$$

$$i_c := i_q - \frac{\left(1 - i_q\right)}{N_c \cdot \tan\left(\phi'_{k2}\right)} = 0.998 \qquad P_{adm3} := \frac{R_d}{A'_3}$$

$$P_{adm3} := c'_{d2} \cdot N_c \cdot b_c \cdot s_c \cdot i_c + q' \cdot N_q \cdot b_q \cdot s_q \cdot i_q + 0.5 \cdot \gamma' \cdot B'_3 \cdot N_\gamma \cdot b_\gamma \cdot s_\gamma \cdot i_\gamma = 449.214\ \frac{\textit{kN}}{\textit{m}^2}$$

$$\frac{V_{d3}}{A'_3} = 34.448\ \frac{\textit{kN}}{\textit{m}^2} \qquad P_{adm3} > \frac{V_{d3}}{A'_3}$$

VERIFICĂ

VERIFICAREA STRUCTURALĂ A FIE - ARMAREA TĂLPII

$L_{af3} := 6\ \textit{m}$

$H_{tot3} := 1.25\ \textit{m}$

$B_3 = 2\ \textit{m}$

$L_3 = 2.5\ \textit{m}$

$\gamma_{BCA} := 16\ \frac{\textit{kN}}{\textit{m}^3}$

$\gamma_{b.a} := 24\ \frac{\textit{kN}}{\textit{m}^3}$

$\gamma_{med} := 22\ \frac{\textit{kN}}{\textit{m}^3}$

$N_{s3} = 6.426\ \textit{kN}$

$M_{xaf3} := M_{x3} + T_{y3} \cdot H_3 = 1.558\ \textit{kN} \cdot \textit{m}$

$M_{yaf3} := M_{y2} + T_{x2} \cdot H_3 = 1.189\ \textit{kN} \cdot \textit{m}$

$G_{f3} := 77.517\ kN$

$V_{d3} := N_{s3} + G_{f3} = 83.943\ kN$

$$P_{13} := \frac{V_{d3}}{L_3 \cdot B_3} + \frac{6 \cdot M_{xaf3}}{{L_3}^2 \cdot B_3} = 17.537\ \frac{kN}{m^2}$$

$$P_{23} := \frac{V_{d3}}{L_3 \cdot B_3} - \frac{6 \cdot M_{xaf3}}{{L_3}^2 \cdot B_3} = 16.041\ \frac{kN}{m^2}$$

$P_{w1} := P_{13}$ $w_1 := 65\ cm$ $w_2 := 52.5\ cm$

$$M_{13} := \left(\left(P_{w1} \cdot w_1 \cdot \frac{w_1}{2}\right) + \frac{\left(P_{13} - P_{w1}\right) \cdot w_1}{2} \cdot \frac{2}{3} \cdot w_1\right) \cdot B_3 = 7.409\ kN \cdot m$$

$$P_{med3} := \frac{\left(P_{13} + P_{23}\right)}{2} = 16.789\ \frac{kN}{m^2}$$

$$M_{23} := \left(P_{med3} \cdot w_2 \cdot \frac{w_2}{2}\right) \cdot L_2 = 5.784\ kN \cdot m$$

MARCA 3 **CLASĂ BETON C16/20** **OȚEL S500**

$f_{ck} := 16\ MPa$ **Strat acoperire beton** $c_{nom} := 4.5\ cm$

$$f_{cd} := \frac{f_{ck}}{1.5} = 10.667\ MPa$$

$f_{yk} := 500\ MPa$

$$f_{yd} := \frac{f_{yk}}{1.15} = 434.783\ MPa$$

$M_{ed} := M_{13}$

$d_3 := c_{nom} + 1\ cm = 0.055\ m$

$d := H_3 - c_{nom} + 10\ cm = 0.905\ m$

$$\mu := \frac{M_{ed}}{L_3 \cdot d^2 \cdot f_{cd}} = 3.392 \cdot 10^{-4} \Rightarrow \quad \mu \le \mu_{lim}\ (0.372)$$

$$w := 1 - \sqrt{1 - 2 \cdot \mu} = 3.393 \cdot 10^{-4}$$

$$A_{nec} := w \cdot L_3 \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.188 \ cm^2$$

$$A_{s.min} := 0.001 \cdot H_3 \cdot B_3 = 17 \ cm^2$$

$A_{seff3} := 8.635 \ cm^2$ **ALEG 11fi10/150**

MARCA 4 **CLASĂ BETON C16/20** **OȚEL S500**

$M_{ed3} := M_{23}$ **Strat acoperire beton** $c_{nom} := 4.5 \ cm$

$H_3 := 0.55 \ m$

$$\mu := \frac{M_{ed3}}{B_3 \cdot d^2 \cdot f_{cd}} = 3.31 \cdot 10^{-4} \quad => \quad \mu \leq \mu_{lim}(0.372)$$

$$w_3 := 1 - \sqrt{1 - 2 \cdot \mu} = 3.311 \cdot 10^{-4}$$

$$A_{nec} := w_3 \cdot B_3 \cdot d \cdot \frac{f_{cd}}{f_{yd}} = 0.147 \ cm^2$$

$$A_{s.min} := 0.001 \cdot h_{c2} \cdot L_3 = 7.5 \ cm^2$$

$A_{seff2} := 5.49 \ cm^2$ **ALEG 7fi10/240**

VERIFICARE LA TASARE

$S_{Ed} \leq S_{Cd}$ $s_{adm} := 80 \ mm$

$S_{Cd} := s_{adm}$

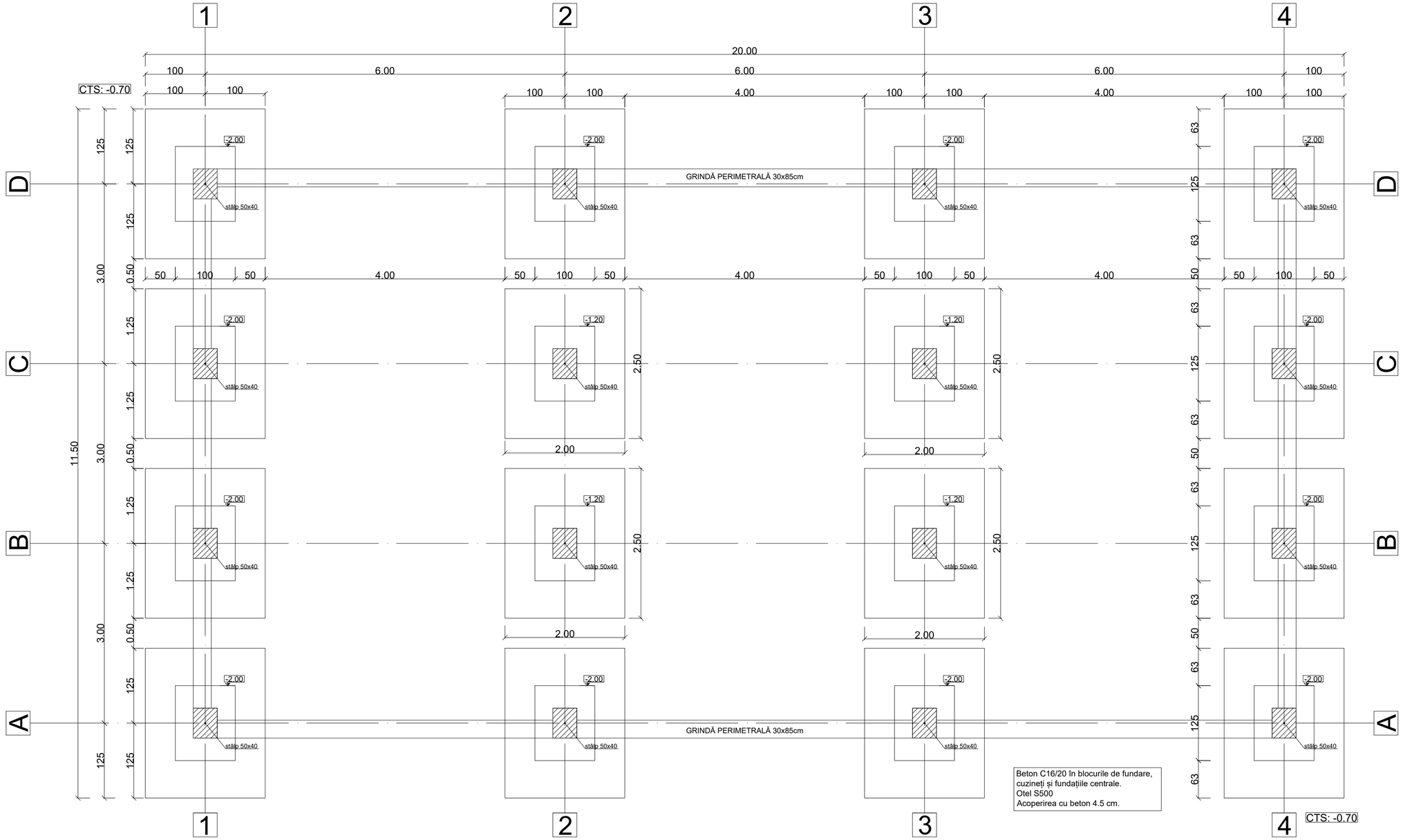
$$P_{net2} := \frac{V_{d2}}{A'_2} - D_{f2} \cdot \gamma_{k2} = 78.595 \ kPa$$

$$P_{net3} := \frac{V_{d3}}{A'_3} - D_{f3} \cdot \gamma_{k2} = -3.538 \ kPa$$

$S_{Ed2} := 0.0346563786 \ m = 34.656 \ mm$ $S_{Ed2} < S_{Cd} = 1$

$S_{Ed3} := 0.03645053672 \ m = 36.451 \ mm$ $S_{Ed3} < S_{Cd} = 1$

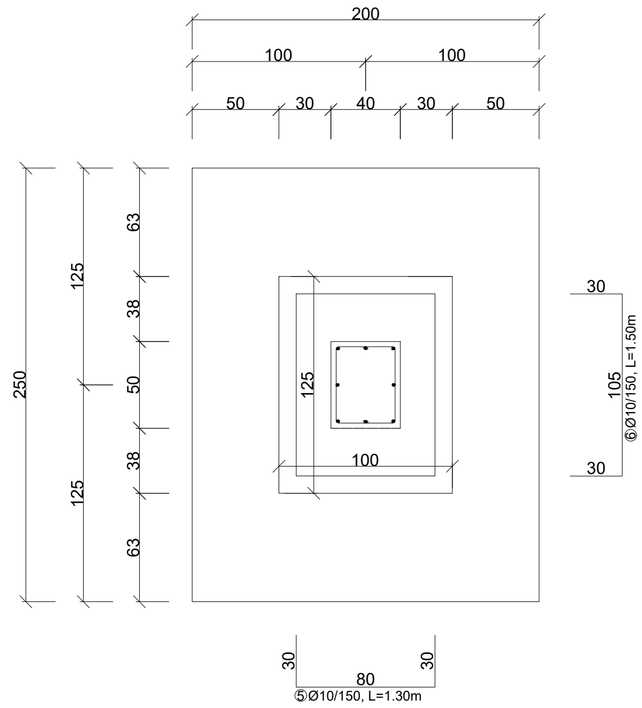
VERIFICĂ



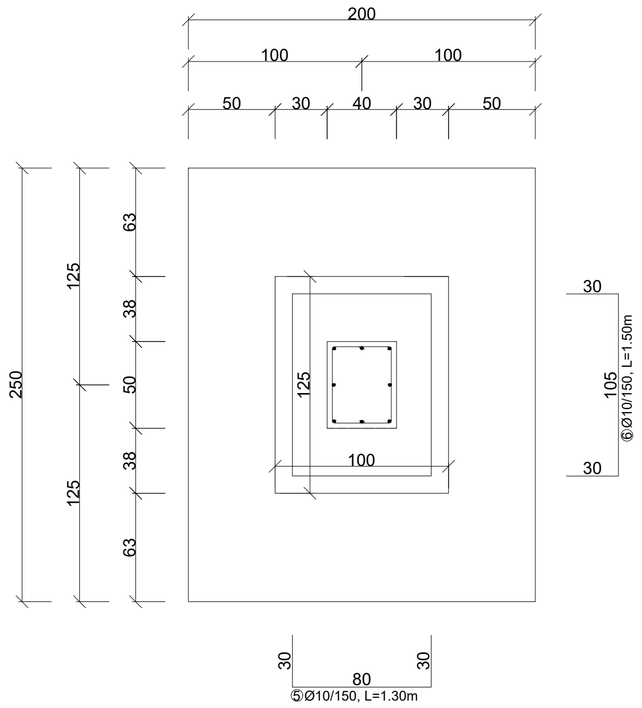
Beton C16/20 în blocurile de fundare, cuzineți și fundațiile centrale.
Otel S500
Acoperirea cu beton 4.5 cm.

UNIVERSITATEA TEHNICĂ CLUJ-NAPOCA FACULTATEA DE CONSTRUCȚII			FUNDAȚII ANUL IV / SECȚIA IUDR (3641) AN UNIVERSITAR 2024-2025		PROIECT 1/2025
	NUME	SEMNĂTURĂ	SCARA 1:50	PLAN FUNDAȚII	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU		DATA 14.01.2025		NR. PLANȘĂ 1
VERIFICAT					

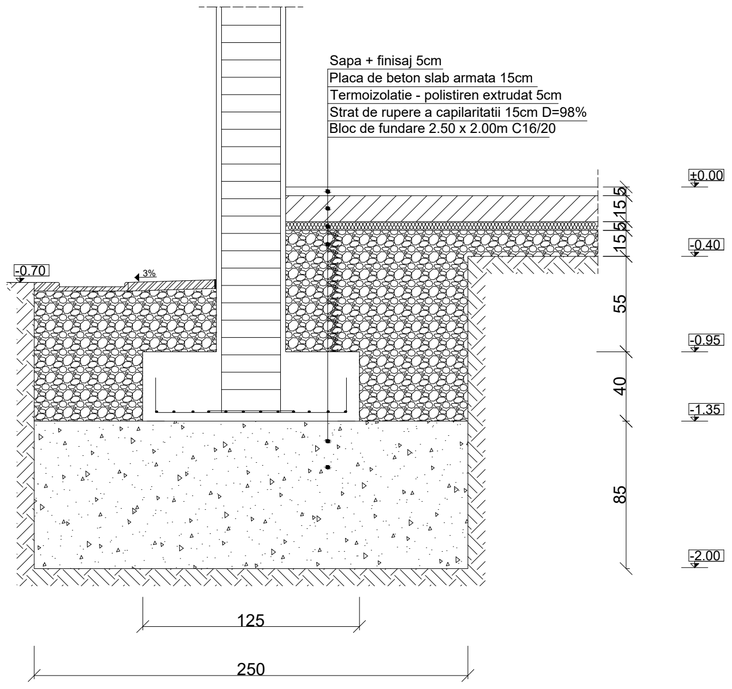
Plan armare și dispunere mustăți stâlp în cuzinet Ax 1/A
scara 1:20



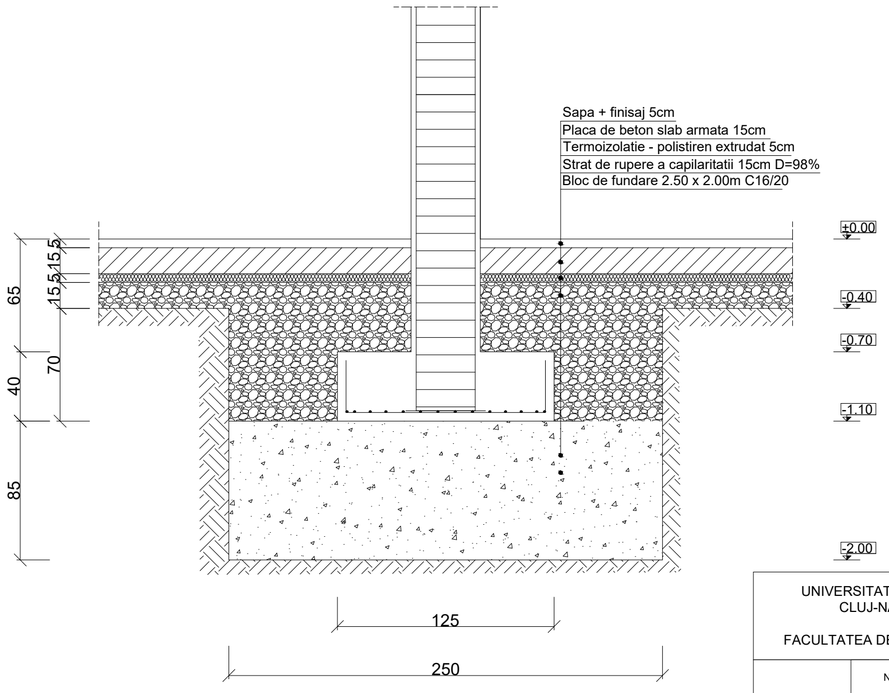
Plan armare și dispunere mustăți stâlp în cuzinet Ax 2/A
scara 1:20



Secțiune transversală fundație Ax 1/A
scara 1:20



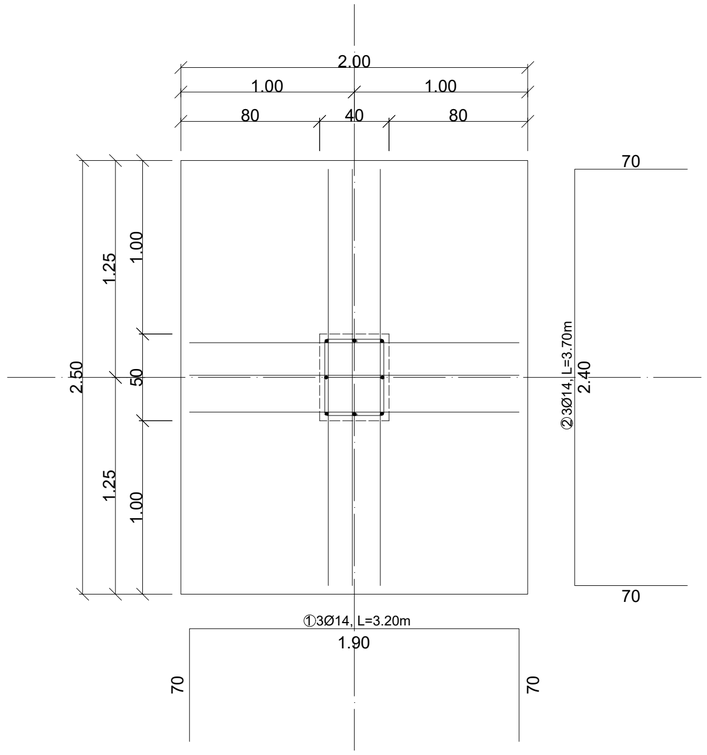
Secțiune transversală fundație Ax 2/A
scara 1:20



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	NUME	SEMNĂTURĂ	SCARA 1:20	DETALII CARACTERISTICE	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU		DATA 14.01.2025		NR. PLANȘĂ 2
VERIFICAT					

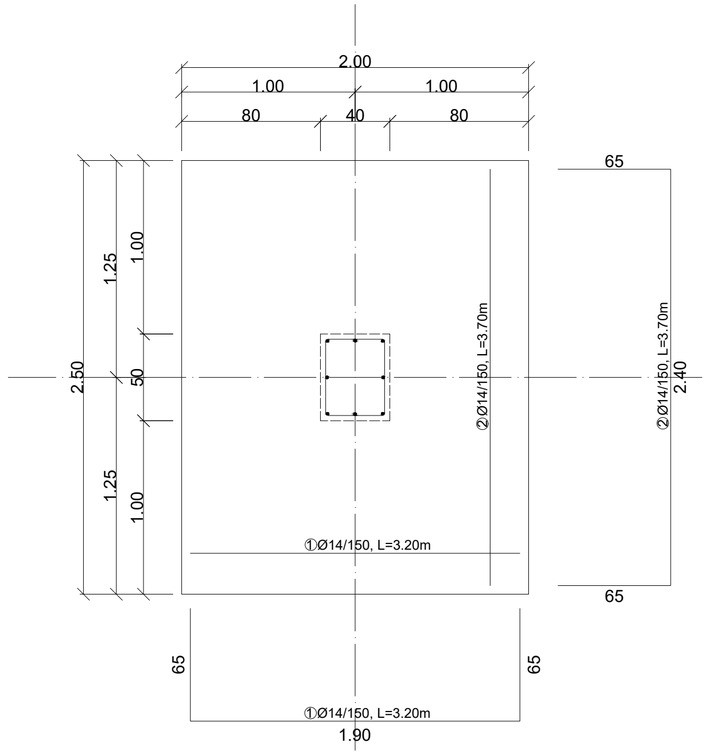
Plan armare superioară și dispunere mustăți stâlp în fundație Ax 2/C

scara 1:20



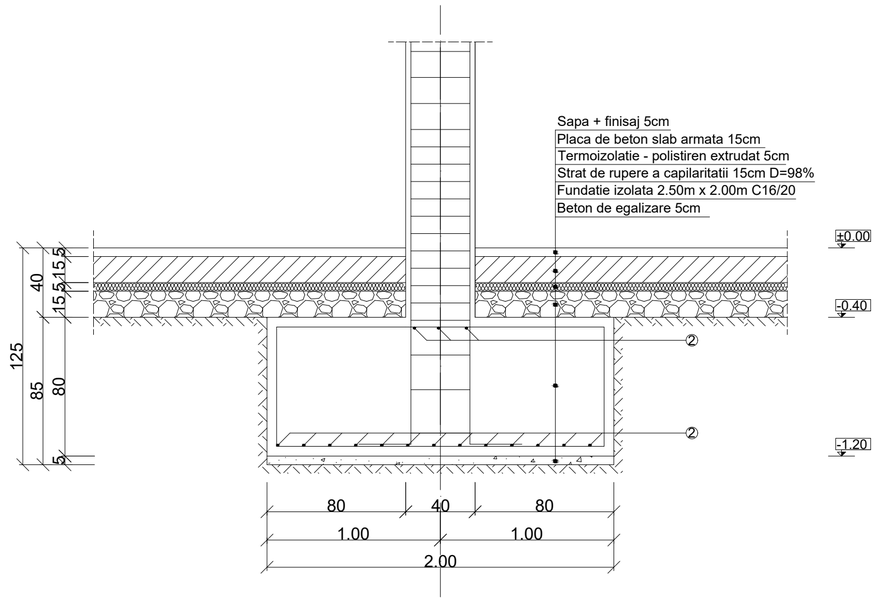
Plan armare inferioară fundație Ax 2/C

scara 1:20



Secțiune transversală fundație Ax 2/C

scara 1:20



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	NUME	SEMNĂTURĂ	SCARA 1:20	DETALII CARACTERISTICE	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU				
VERIFICAT			DATA 14.01.2025		NR. PLANȘĂ 3